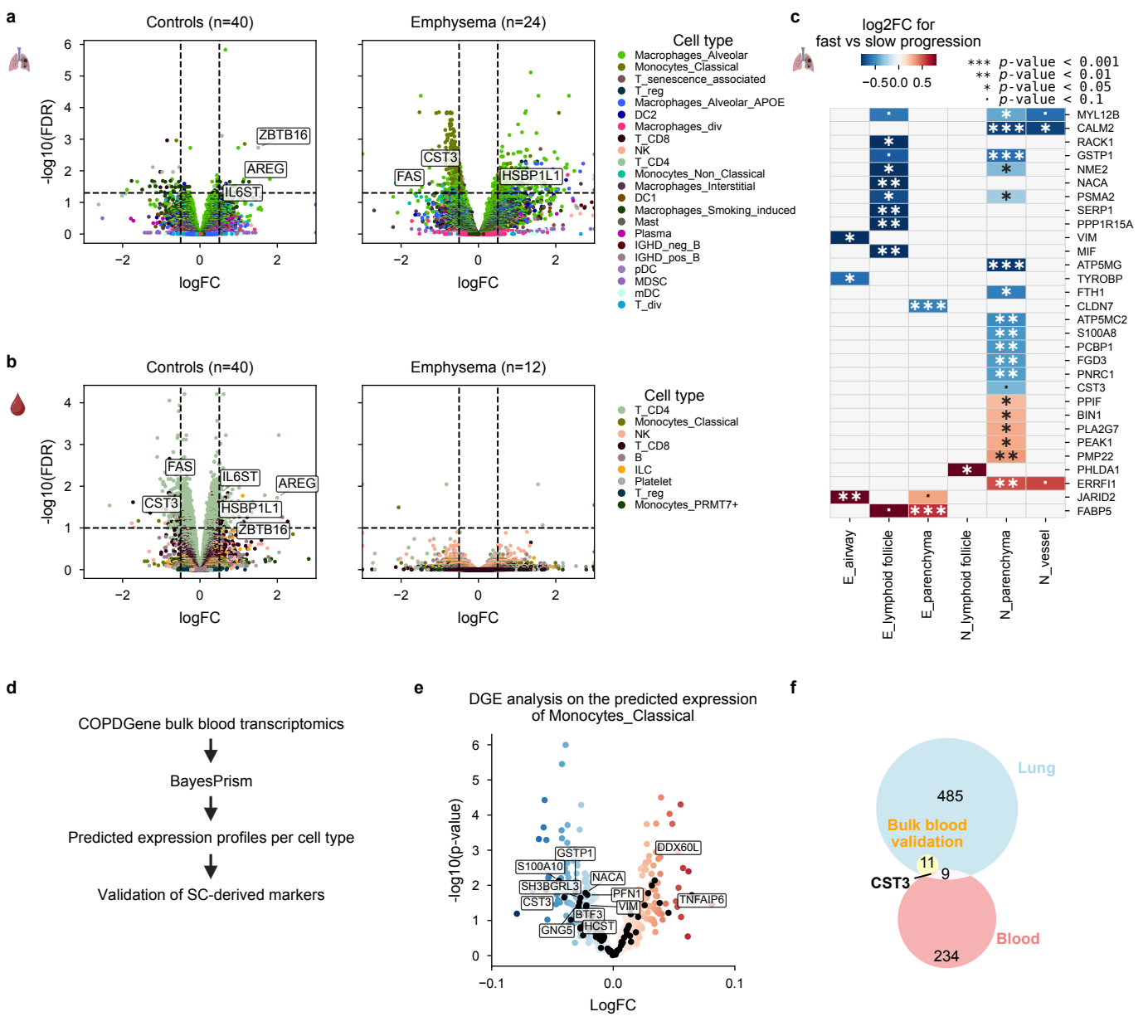

a
Controls (n=40)
Emphysema (n=24)
-log10(FDR)
logFC
ZBTB16
AREG
IL6ST
CST3
FAS
HSBP1L1
Cell type
Macrophages_Alveolar
Monocytes_Classical
T_senescence_associated
T_reg
Macrophages_Alveolar_APOE
DC2
Macrophages_div
T_CD8
NK
T_CD4
Monocytes_Non_Classical
Macrophages_Interstitial
DC1
Macrophages_Smoking_induced
Mast
Plasma
IGHD_neg_B
IGHD_pos_B
pDC
MDSC
mDC
T_div
b
Controls (n=40)
Emphysema (n=12)
-log10(FDR)
logFC
FAS
IL6ST
AREG
CST3
HSBP1L1
ZBTB16
Cell type
T_CD4
Monocytes_Classical
NK
T_CD8
B
ILC
Platelet
T_reg
Monocytes_PRMT7+
c
log2FC for
fast vs slow progression
−0.5 0.0 0.5
*** p-value < 0.001
** p-value < 0.01
* p-value < 0.05
· p-value < 0.1
MYL12B
CALM2
RACK1
GSTP1
NME2
NACA
PSMA2
SERP1
PPP1R15A
VIM
MIF
ATP5MG
TYROBP
FTH1
CLDN7
ATP5MC2
S100A8
PCBP1
FGD3
PNRC1
CST3
PPIF
BIN1
PLA2G7
PEAK1
PMP22
PHLDA1
ERRFI1
JARID2
FABP5
E_airway
E_lymphoid follicle
E_parenchyma
N_lymphoid follicle
N_parenchyma
N_vessel
d
COPDGene bulk blood transcriptomics
BayesPrism
Predicted expression profiles per cell type
Validation of SC-derived markers
e
DGE analysis on the predicted expression
of Monocytes_Classical
-log10(p-value)
LogFC
GSTP1
S100A10
SH3BGRL3
CST3
BTF3
GNG5
HCST
NACA
PFN1
VIM
DDX60L
TNFAIP6
f
Lung
485
Bulk blood
validation
11
CST3
9
Blood
234